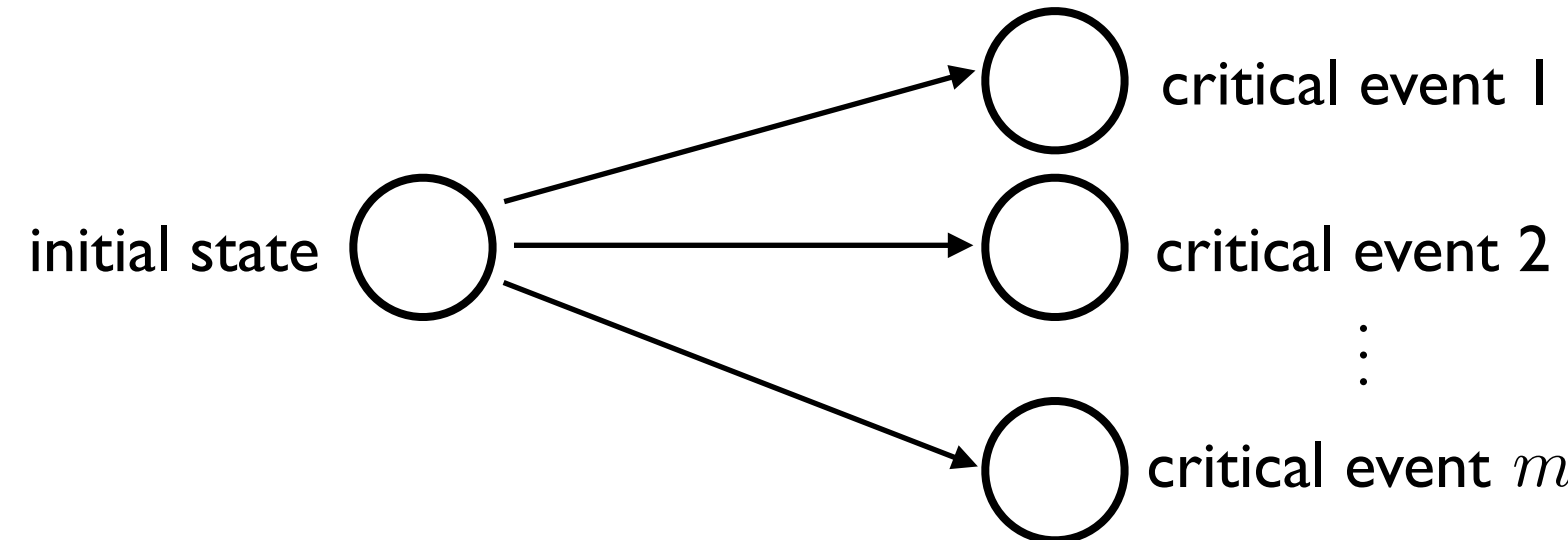


Background & Motivation

Competing Risks in Survival Analysis

- Patients face multiple potential events, and the occurrence of one precludes the others
- What critical event will happen earliest, and when?



Cumulative Incidence Function (CIF)

$$F_{\delta}(t|x) \triangleq \mathbb{P}(\Delta^* = \delta, T \leq t | X = x) \quad \delta = 1, \dots, m, t \geq 0$$

Earliest event is δ and it happens by time t for feature vector x

Problem

- Recently developed competing risks models are black-box & not inherently interpretable (e.g., DeepHit (2018), Neural Fine-Gray (2023), SurvivalBoost (2025))

Goal

- Develop a deep competing risks model that is interpretable

TL;DR

We propose a conditional Aalen-Johansen estimator that learns a kernel function to represent patients as **weighted combinations of clusters**, where each cluster has a known, interpretable risk profile

Classic Aalen-Johansen Estimator (1978)

Key Concept:

Calculate risk by counting observed events at specific times (focus on competing risks & not general multi-state processes)

Steps:

- Find all unique times in which any critical event occurred

$$0 < t_1 < t_2 < \dots < t_L \quad L = \# \text{ unique times of any critical event}$$

- At each event time, count

$$(i) \# \text{ times an event occurred } d_{\delta,\ell} \triangleq \sum_{j=1}^n \mathbb{1}\{\Delta_j = \delta, Y_j = t_{\ell}\}$$

$$(ii) \# \text{ patients at risk } n_{\ell} \triangleq \sum_{j=1}^n \mathbb{1}\{Y_j \geq t_{\ell}\}$$

- Output

$$\hat{F}_{\delta}^{AJ}(t) \triangleq \sum_{\ell \in \{1, \dots, L\} \text{ s.t. } t_{\ell} \leq t} \hat{S}^{KM}(t_{\ell-1}) \frac{d_{\delta,\ell}}{n_{\ell}}$$

Survival prob. just before t_{ℓ} (Kaplan-Meier)

$$\hat{S}^{KM}(t_{\ell-1}) \triangleq \prod_{\ell \in \{1, \dots, L\} \text{ s.t. } t_{\ell} \leq t} \left(1 - \frac{\sum_{\delta=1}^m d_{\delta,\ell}}{n_{\ell}}\right)$$

Probability of event δ at t_{ℓ}

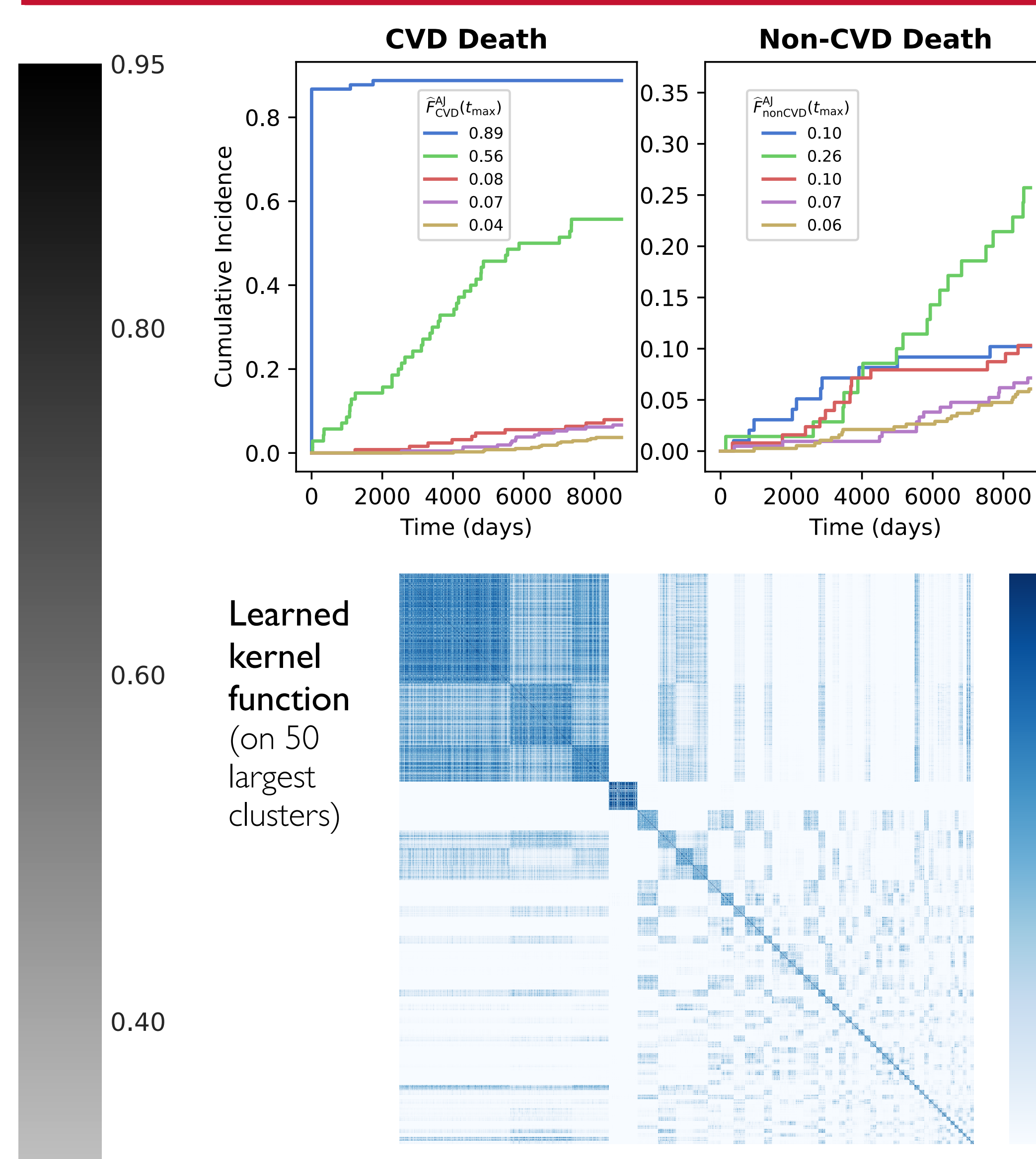
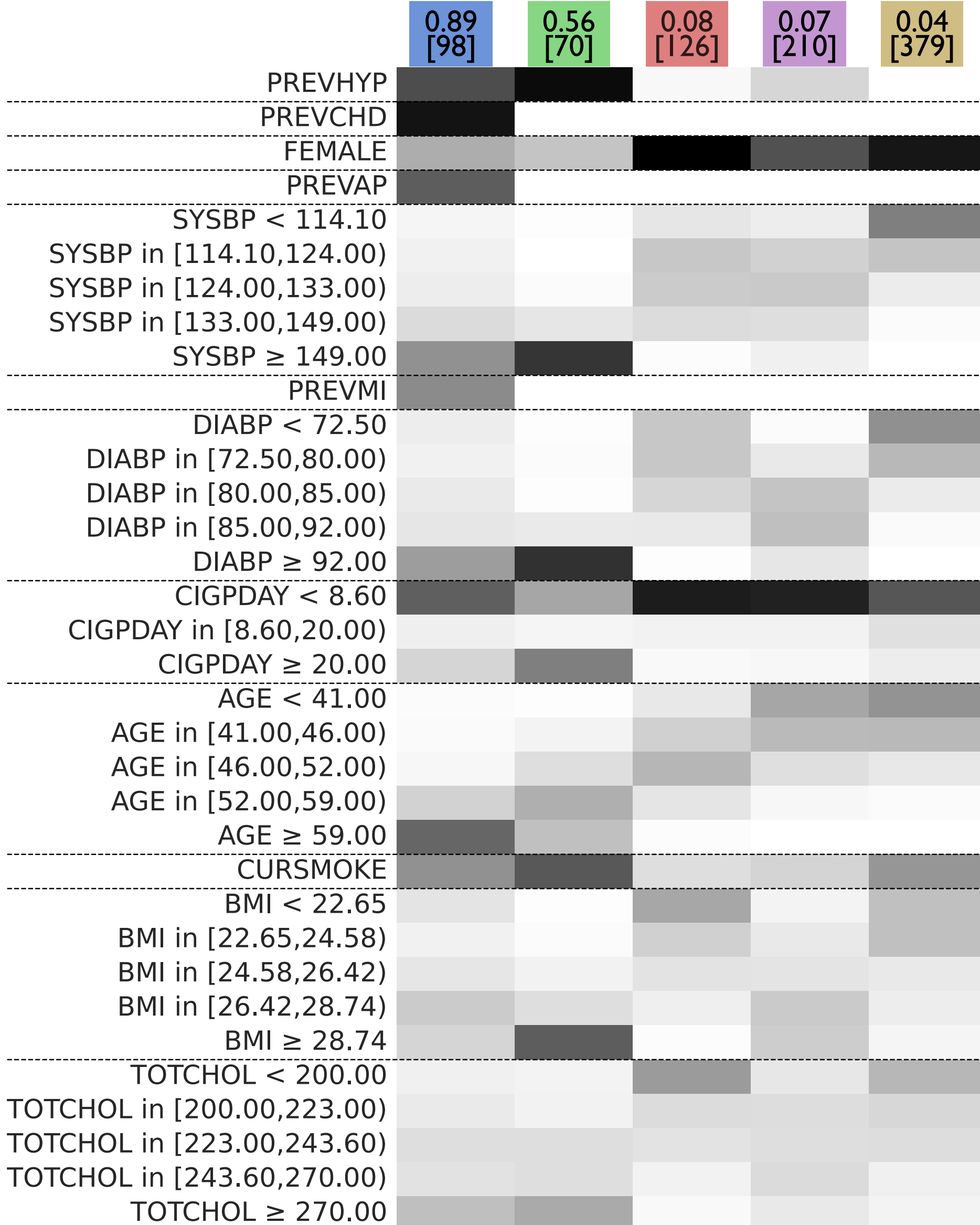
Limitation:

A population-level estimator; doesn't handle individual covariates

Model Interpretation

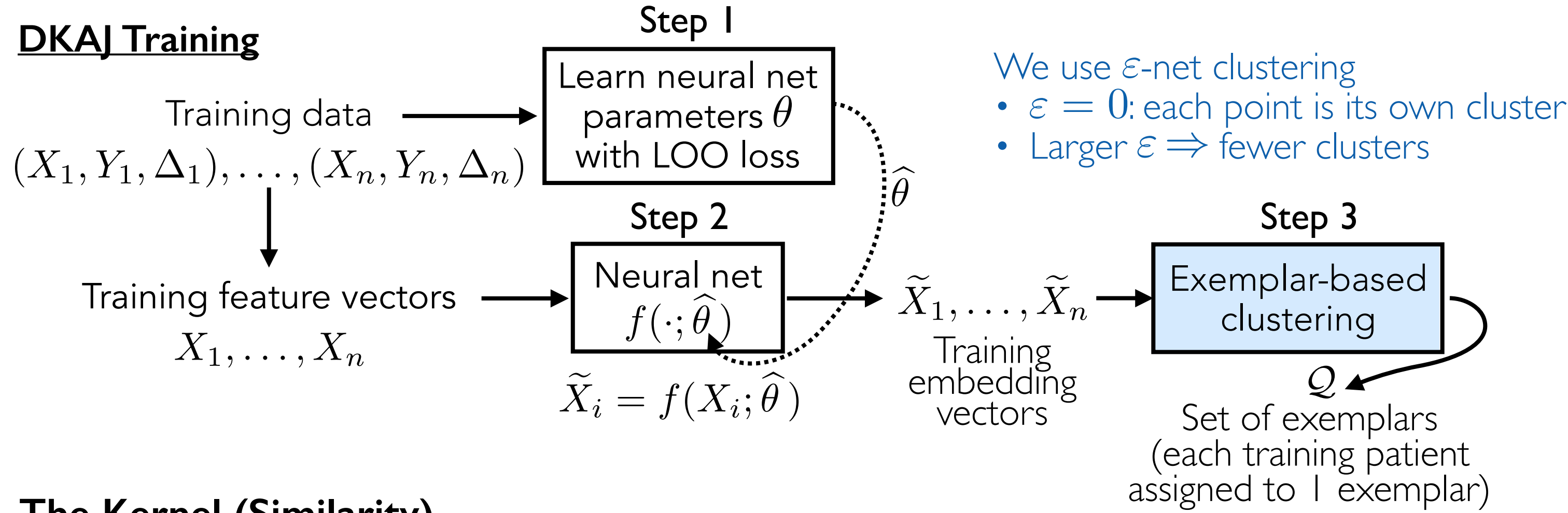
Cluster-level

Largest 5 clusters, sorted by CVD CIF at max obs time; cluster sizes are stated in square brackets



Deep Kernel Aalen-Johansen Estimator

DKAJ Training



The Kernel (Similarity)

How to measure similarity in the embedding space? Use a kernel function parameterized by a neural net:

$$K(x, x') \triangleq \exp(-\|f(x; \theta) - f(x'; \theta)\|^2)$$

Leave-One-Out Loss

$$\text{loss} = -\log \mathcal{L} = -\log \left\{ \prod_{i=1}^n (\lambda_{\Delta_i}(Y_i|X_i))^{\mathbb{1}\{\Delta_i \neq 0\}} S(Y_i|X_i) \right\}$$

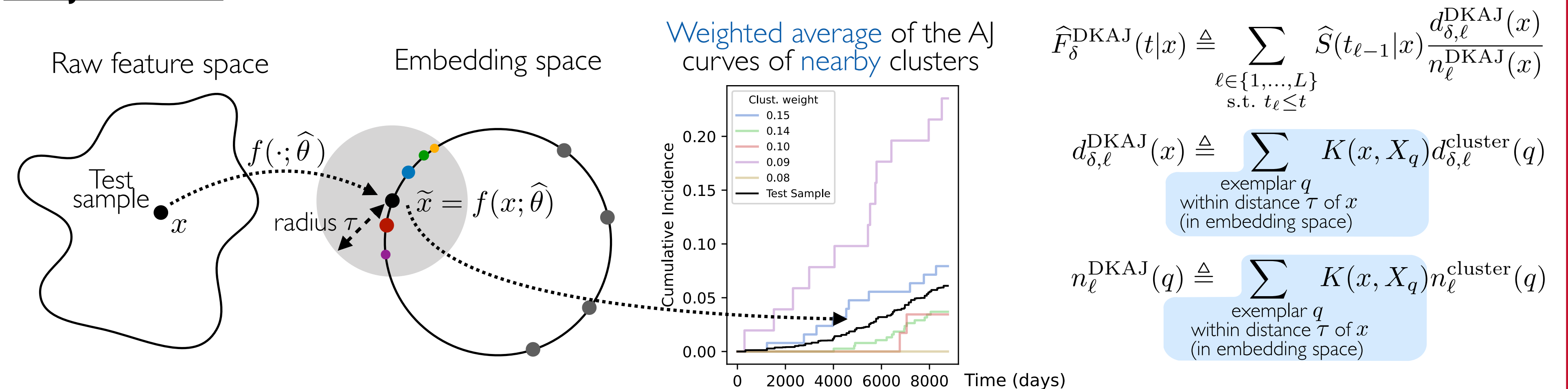
$$\text{LOO loss} = \sum_{i=1}^n \sum_{\delta=1}^m \left[-\mathbb{1}\{\Delta_i = \delta\} \log \psi_{\delta, \kappa(Y_i)}^{\neg i}(X_i; \theta) + \sum_{\ell=1}^L \psi_{\delta, \ell}^{\neg i}(X_i; \theta) \right] + \text{constant}$$

Problem: i -th training point's loss function uses hazard function prediction that has access to the i -th training point's ground truth

$$\psi_{\delta, \ell}^{\neg i}(x; \theta) = \frac{\sum_{j \neq i} \mathbb{1}\{\Delta_j = \delta, Y_j = t_{\ell}\} \exp(-\|f(x; \theta) - f(X_j; \theta)\|^2)}{\sum_{j \neq i} \mathbb{1}\{Y_j \geq t_{\ell}\} \exp(-\|f(x; \theta) - f(X_j; \theta)\|^2)}$$

Discrete time index corresponding to Y_i

DKAJ Prediction

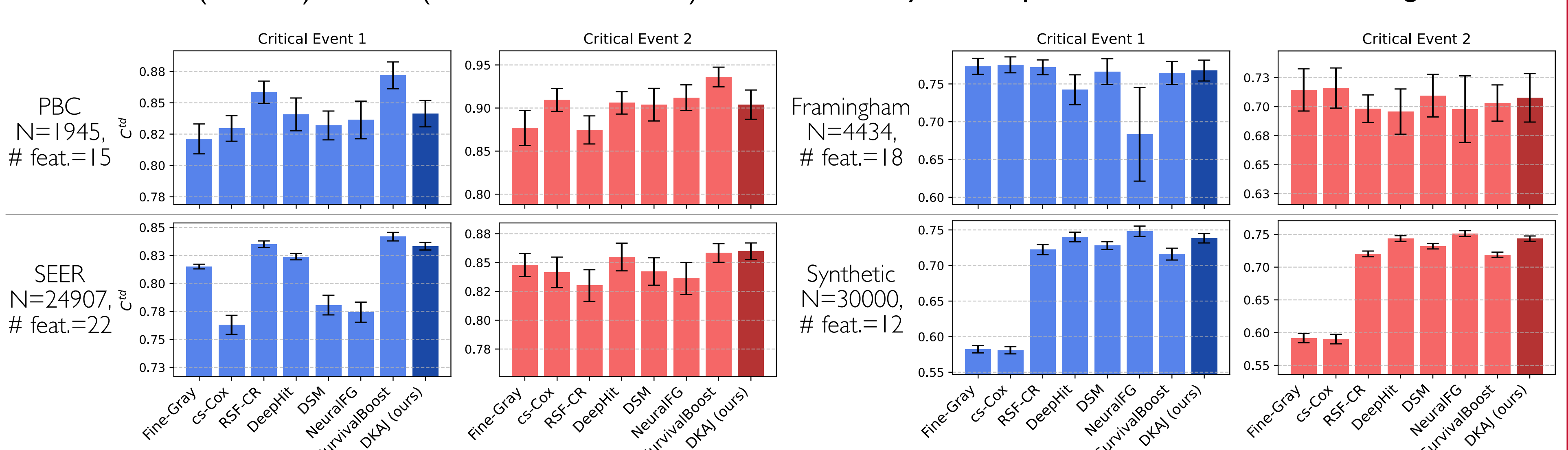


Experiments

Setup & Protocol

- Data: 4 standard competing risks datasets
- Split: 10 random train/test splits (70% / 30%)
- Metrics: C^{td} (discrim.) and IBS (calibration + discrim.)

- ✓ **Ablation**: validated importance of LOO loss & TUNA init
- ✓ **Computation cost**: similar scale as NeuralFG/SurvivalBoost
- ✓ **Scalability**: robust performance on smaller training sets



Conclusions & Future Work

Conclusions & Impact

- High-performing, interpretable deep competing risks framework
- Generalizes the Survival Kernets (Chen, 2024) framework to the competing risks setting using the Aalen-Johansen estimator
- Transparent predictions via weighted classical estimator
- Enables cluster-level & individual-level visualization
- Allows probing of learned kernel structure

Limitations & Extensions

- Modifications**: Explore training with proper scoring rule loss, alternative clustering strategies, other kernel functions (e.g., time-dependent, event-specific)
- Theory**: Establish error upper bound on predicted CIFs
- Scope**: Generalize to multi-state processes (the general setting AJ was designed for)